Pakistan Journal of Pharmacol. ISSN 0255-7088
Vol.16(1), January 1999, pp.9-11

ANTIBACTERIAL AND KINETIC STUDIES ON SOME CATIONIC SURFACTANTS

FOUZIA HASSAN*, ANWER EJAZ BEG, S. BAQIR SHYUM NAQVI, AAMIR MEHDI AND HARRIS SHOAIB

**Department of Pharmaceutics, Faculty of Pharmacy*
University of Karachi, Karachi-75270

ABSTRACT

The antibacterial activity of four structurally related cationic surfactants were determined against ten different species of bacterial cultures. From the study it is revealed that three surfactants out of four show some level of antibacterial activity against all the Gram positive and Gram negative bacteria used in the study, while one of the surfactant was not effective against the tested organisms. The above surfactants were also tested for their kinetic effect.

INTRODUCTION

The use of surfactants as Emulsifiers, solubilizers, wetting agent, preservative and antibacterial is well documented in the literature [Vindevoghel et al. (1994), Macall, Patrick (1993), Dowell (1993), Nakamuo (1996), Ochoa (1996), Mukerjee (1995), Saraf (1995), Allen (1995), Nicheal (1995), Suzuki (1996)]. However antibacterial activity of surfactants has been reported to a limited extent. The present investigation is a key step in this direction. The main objective of this study is to evaluate the comparative effects of cationic surfactants on hydrolysis of ester as well as their antibacterial activity.

MATERIAL AND METHOD

Surfactants used were dodecyl, tetradecyl, hexadecyl trimethyl ammonium brômide and cetyl Pyridinium bromide supplied by sigma lab. Ester used was paranirophenyl propionate (PNPP). Water used in the study was double distilled. Buffer used was Delory & Kings (Docomenta G., 1962) carbonate bicarbonate buffer. Spectrophotometer 160 UV was used for kinetic study. Kinetic study was performed as described elsewhere (Hassan F. 1996). Ten clinical isolates were collected from pathological labs. And their sensitivity was determined along with the standards Table 1. These were further identified by morphological and biochemical methods. Cultures were grown in Muller-Hinton broth (MHB) at 37°C for 2-8 hours. Turbidity was matched with that of Mac Farland standard 0.5 (Bertina B.W., 1987). Sensitivity test to different cationic surfactants was carried out by Agar Dilution method as described by (Lorian V. 1991). The results were interpreted by observing the plates for complete inhibition of growth.

RESULT AND DISCUSSION

Kinetic study:

Form the previous study (Hassan F., 1996) it is revealed that all cationic surfactants accelerate the rate of hydrolysis of the ester above their critical micelle concentration than non surfactant system. Although the magnitude of the hydrolysis rate being different in these surfactants Micellar catalysis tends to be more pronounced for surfactants having longer alkyl chain i.e. for those that are more hydrophobic as is apparent in the study (Hassan F., 1996). The effect of chain length of sodium alkyl sulphate on the acid catalyzed hydrolysis by methyl or-tho benzoate is also observed in the study of Dunalp & Cordes 1968, which were to ascribed by some chemical change in micelle structure. The catalytic efficiency of the different alkyl group is in the order hexadecyl > tetradecyl > dodecyl> octyl. Such effect can be rationalized by postulating increased charge density of the ions of the surfactant of micelle with increasing carbon chain length which in turn would increase the electric field of the reaction site.

Antibacterial activity:

The antibacterial activity of these compounds was tested against ten different Gram positive and Gram negative bacteria as shown in Table-1. From the present communication it is indicated Table-2 that cationic surfactants having alkyl chain length between C_{12}, C_{14}, C_{16} show promising antibacterial activity, where as compound having alkyl chain length C_{20} does not show any antibacterial activity.

Table-1

Organism used in the study

Gram Positive		Gram Negative	
Staphlococcus aureus	Clinical Isolate	Salmonellapara A	Clinical Isolate
Staphlococcus aureus	" "	Standard of S. typhi	786 NTCC
Staphlococcus aureus	" "	Echericha coli	9001 ATCC
Staphlococcus aureus	" "	Pseudomonas areginosa	Clinical Isolate
Standard of aureus	2365 ATCC		
Streptococcus pyogens	Clinical Isolate		

Table-2
Antibacteial activity of cationic surfactants

Name of Organism	Dodecyl trimethyl ammonium bromide	Tetradecyl trimethyl bromide	Hexeadecyl trimethyl bromide	Cetyl Pyridinm bromide
Staphlococcus aureus	Positive	Positive	Positive	Negative
Staphlococcus aureus	"	"	"	"
Staphlococcus aureus	"	"	"	"
Staphlococcus aureus	"	"	"	"
Standard of A. Pyogens	"	"	"	"
Salmonella Para A	"	"	"	"
Standard of S. typhi	"	"	"	"
Echericha coli	"	"	"	"
Pseudomonos areginosa	"	"	"	"

REFERENCES

Allen A., George L. (1995). *PCT Int. App. Wo aj.* **35**: 364.
Bertina B.W. (1987). Diagnostic Procedures for Bacterial Infections. 7th edn. American Public Health Association, p.615.
Vindevoghel Ph; Noguvès P., Guyot A. (1994). *J. Appl. Polym. Sci.*52(13): 1879.
Dowell T.J. (1993). Eurpal Apple P 562, 639.
Macall P.C. (1993). PCt 2nd App wo 9307, 545.
Micheals Ed.B. (1995). *C.A.* Vol.122, No.14.
Mukarjee P. Shema R., Pyter R.A., Gumkowski M.J. (1995). ACS Symp., Ser., 615.
Nakamuo T. K., Hiroyaki (1996). Jpn Kokai TokkyoJP08, 02:059.
Ochoa et al. (1996). *Biochem., Biopcys Acta,* **1240**(2): 209.
Saraf Y., Bhgwat S.S. (1995). *Sep. Technol.* **5**(4): 207.
Suzuki T., Yokama K. (1996). *C.A.* Vol.124.